

KADENA AB, Japan

WMO: 479310

Lat: 26.356N

Lon: 127.768E

Elev: 47

StdP: 100.77

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 42204

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.9	10.2	2.8	4.7	12.9	3.9	5.0	13.4	11.0	15.6	10.2	15.6	3.0	60	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	33.1	26.9	32.2	26.9	31.8	27.0	29.3	30.5	28.7	30.2	28.2	29.8	5.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.1	26.0	29.9	28.3	24.7	29.4	27.9	24.1	29.1	97.0	30.5	93.8	30.4	91.1	29.5	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	10.0	8.9	DB	6.0	33.9	1.7	1.0	4.8	34.6	3.8	35.2	2.8	35.8	1.6	36.5
			WB	4.8	29.5	1.2	1.1	4.0	30.3	3.3	30.9	2.6	31.6	1.8	32.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	16.3	16.8	18.3	21.0	23.9	27.1	29.1	28.9	27.7	25.0	21.7	18.0	(6)
(7)		DBStd	5.12	2.62	2.83	3.02	2.76	2.27	2.05	1.13	1.20	1.57	2.13	2.44	2.65	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	225	75	61	41	7	0	0	0	0	0	0	3	39	(9)
(10)		CDD10.0	4684	195	189	256	331	431	513	592	586	529	464	352	248	(10)
(11)		CDD18.3	1868	12	17	38	88	172	263	333	327	279	206	104	29	(11)
(12)		CDH23.3	17472	11	20	70	328	1069	2684	4136	4029	3066	1577	431	50	(12)
(13)		CDH26.7	5836	0	0	3	31	159	827	1736	1658	1047	341	33	1	(13)
(14)	Wind	WSAvg	4.7	4.6	4.6	4.6	4.5	4.3	4.9	4.7	4.7	4.9	4.7	4.6	(14)	
(15)	Precipitation	PrecAvg	2078	121	110	153	165	244	256	191	248	181	159	108	139	(15)
(16)		PrecMax	3135	309	406	367	460	691	570	620	651	459	585	260	822	(16)
(17)		PrecMin	1034	27	28	20	21	16	22	12	38	44	4	4	23	(17)
(18)		PrecStd	544	56	83	90	99	140	139	171	148	97	123	75	140	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	25.2	26.9	29.0	30.2	32.9	34.1	34.1	33.1	31.9	28.9	26.2	(19)
(20)		MCWB	19.8	20.8	22.4	24.8	26.1	27.0	27.1	26.6	25.8	24.9	24.1	21.5	(20)	
(21)		2%	DB	23.1	23.9	25.0	27.8	29.2	31.8	33.1	33.1	32.1	30.4	27.8	24.8	(21)
(22)		MCWB	18.9	20.2	21.8	24.0	25.6	27.4	27.1	27.1	26.3	25.5	23.4	20.7	(22)	
(23)		5%	DB	22.0	22.8	24.0	26.2	28.2	31.0	32.2	32.2	31.2	29.4	26.8	23.7	(23)
(24)		MCWB	18.2	19.6	20.7	22.8	25.1	27.5	27.2	27.0	26.4	25.0	22.7	19.6	(24)	
(25)		10%	DB	21.0	21.8	23.0	25.1	27.2	30.2	31.8	31.6	30.8	28.5	25.8	22.7	(25)
(26)		MCWB	17.4	18.8	20.0	22.3	24.4	27.0	27.2	27.0	26.4	24.4	22.1	18.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.0	22.6	23.7	26.2	28.2	30.1	30.2	30.2	29.8	28.9	26.1	23.4	(27)
(28)		MDCB	23.2	24.2	25.1	27.3	28.8	30.7	31.1	31.2	30.5	29.3	27.4	24.8	(28)	
(29)		2%	WB	20.4	21.4	22.6	24.9	26.9	28.9	29.1	29.0	28.6	27.2	25.0	22.0	(29)
(30)		MDCB	22.0	22.9	24.2	26.4	28.0	29.9	30.7	30.6	29.8	28.3	26.1	23.5	(30)	
(31)		5%	WB	19.2	20.4	21.7	24.0	26.0	28.1	28.5	28.3	27.9	26.2	24.1	20.8	(31)
(32)		MDCB	21.2	21.9	23.3	25.7	27.5	29.4	30.3	30.1	29.4	28.2	25.6	22.5	(32)	
(33)		10%	WB	18.0	19.3	20.7	23.1	25.3	27.4	28.0	27.9	27.1	25.5	23.0	19.7	(33)
(34)		MDCB	20.2	21.1	22.4	24.8	26.8	29.2	30.0	29.8	29.2	27.7	25.0	21.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	6.6	6.6	6.8	6.5	5.8	4.8	5.2	5.3	5.8	6.3	6.6	6.7	(35)	
(36)		5% DB	MDCDBR	8.0	7.5	6.9	6.3	5.9	5.1	6.0	6.1	6.2	6.8	7.1	7.8	(36)
(37)		MCWBR	5.4	5.1	4.9	3.9	3.6	2.7	2.5	2.7	3.1	3.6	4.2	4.9	(37)	
(38)		5% WB	MDCDBR	6.2	6.1	6.0	5.4	5.1	4.7	5.0	5.2	5.2	5.4	5.3	6.0	(38)
(39)	MCWBR	5.2	5.1	5.2	4.0	3.8	3.2	2.8	3.0	3.4	3.6	4.0	4.9	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.458	0.487	0.560	0.568	0.513	0.471	0.471	0.488	0.484	0.482	0.465	0.457	(40)	
(41)		taud	2.131	2.068	1.899	1.939	2.161	2.343	2.339	2.278	2.259	2.204	2.207	2.141	(41)	
(42)		Ebn at Noon	775	785	750	757	798	826	826	811	806	782	765	756	(42)	
(43)		Edh at Noon	137	156	195	192	153	127	127	135	134	135	126	130	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.41	3.04	3.77	4.41	4.90	5.42	6.51	5.95	5.21	4.08	2.98	2.44	(44)	
(45)		RadStd	0.29	0.46	0.36	0.43	0.64	0.49	0.53	0.40	0.39	0.35	0.27	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	-1.02	N/A	N/A	+0.51	+0.63	N/A	+50	N/A	N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	+0.41	+0.53	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page